Example 2.79

Evaluate $\int_2^5 \frac{\sqrt{x}}{\sqrt{x}+\sqrt{7-x}}\,dx$

Solution:

Let $I = \int_2^5 \frac{\sqrt{x}}{\sqrt{x}+\sqrt{7-x}}\,dx \quad \ldots(1)$

$$I = \int_2^5 \frac{\sqrt{2+5-x}}{\sqrt{2+5-x}+\sqrt{7-(2+5-x)}}\,dx$$

$$\left[\because \int_a^b f(x)dx = \int_a^b f(a+b-x)dx\right]$$

$$I = \int_2^5 \frac{\sqrt{7-x}}{\sqrt{7-x}+\sqrt{x}}\,dx \ldots(2)$$

$(1)+(2) \Rightarrow$

$$2I = \int_2^5 \left[\frac{\sqrt{x}}{\sqrt{x}+\sqrt{7-x}} + \frac{\sqrt{7-x}}{\sqrt{7-x}+\sqrt{x}}\right]dx$$

$$= \int_2^5 \left[\frac{\sqrt{x}+\sqrt{7-x}}{\sqrt{x}+\sqrt{7-x}}\right]dx$$

$$= \int_2^5 dx = [x]_2^5 = 3$$

$$\therefore \quad I = \frac{3}{2}$$

Exercise 2.9

Evaluate the following using properties of definite integrals:

1. $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} x^3 \cos^3 x\,dx$
2. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2\theta\,d\theta$
3. $\int_{-1}^{1} \log\left(\frac{2-x}{2+x}\right)dx$
4. $\int_0^{\frac{\pi}{2}} \frac{\sin^7 x}{\sin^7 x+\cos^7 x}\,dx$
5. $\int_0^1 \log\left(\frac{1}{x}-1\right)dx$
6. $\int_0^1 \frac{x}{(1-x)^{\frac{3}{4}}}\,dx$

2.2.3 Gamma Integral

Gamma integral is an important result which is very useful in the evaluation of a particular type of an improper definite integrals.

First, let us know about the concepts of indefinite integrals, proper definite integrals and improper definite integrals.

Indefinite integral:

An integral function which is expressed without limits, and so containing an arbitrary constant is called an indefinite integral

Example: $\int e^{-t}dt$

Proper definite integral:

Proper definite integral is an integral function, which has both the limits a and b are finite and the integrand $f(x)$ is continuous in $[a, b]$.

Example: $\int_0^1 e^{-t}dt$

Improper definite integral:

An improper definite integral is an integral function, in which the limits either a or b or both are infinite, or the integrand $f(x)$ becomes infinite at some points of the interval $[a, b]$.

Example: $\int_0^\infty e^{-t}dt$

Definition 2.3

For $n > 0$, $\int_0^\infty x^{n-1}e^{-x}dx$ is known known as Gamma function and is denoted by $\Gamma(n)$ [read as Gamma of n].

Note

If n is a positive integer, then $\int_0^\infty x^n e^{-ax}\,dx = \frac{n!}{a^{n+1}}$ is the particular case of Gamma Integral.

Properties:

1. $\Gamma(n) = (n-1)\Gamma(n-1),\ n>1$
2. $\Gamma(n+1) = n\Gamma(n),\ n>0$

XII Std - Business Maths & Stat EM Chapter 2.indd 50 2/28/2020 5:41:05 PM